



Interpretation of Cerebrospinal Fluid White Blood Cell Counts in Young Infants With a Traumatic Lumbar Puncture

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Study objective: We determine the optimal correction factor for cerebrospinal fluid WBC counts in infants with traumatic lumbar punctures.

Methods: We performed a secondary analysis of a retrospective cohort of infants aged 60 days or younger and with a traumatic lumbar puncture (cerebrospinal fluid RBC count $\geq 10,000$ cells/mm³) at 20 participating centers. Cerebrospinal fluid pleocytosis was defined as a cerebrospinal fluid WBC count greater than or equal to 20 cells/mm³ for infants aged 28 days or younger and greater than or equal to 10 cells/mm³ for infants aged 29 to 60 days; bacterial meningitis was defined as growth of pathogenic bacteria from cerebrospinal fluid culture. Using linear regression, we derived a cerebrospinal fluid WBC correction factor and compared the uncorrected with the corrected cerebrospinal fluid WBC count for the detection of bacterial meningitis.

Results: Of the eligible 20,319 lumbar punctures, 2,880 (14%) were traumatic, and 33 of these patients (1.1%) had bacterial meningitis. The derived cerebrospinal fluid RBCs:WBCs ratio was 877:1 (95% confidence interval [CI] 805 to 961:1). Compared with the uncorrected cerebrospinal fluid WBC count, the corrected one had lower sensitivity for bacterial meningitis (88% uncorrected versus 67% corrected; difference 21%; 95% CI 10% to 37%) but resulted in fewer infants with cerebrospinal fluid pleocytosis (78% uncorrected versus 33% corrected; difference 45%; 95% CI 43% to 47%). Cerebrospinal fluid WBC count correction resulted in the misclassification of 7 additional infants with bacterial meningitis, who were misclassified as not having cerebrospinal fluid pleocytosis; only 1 of these infants was older than 28 days.

Conclusion: Correction of the cerebrospinal fluid WBC count substantially reduced the number of infants with cerebrospinal fluid pleocytosis while misclassifying only 1 infant with bacterial meningitis of those aged 29 to 60 days. [Ann Emerg Med. 2017;69:622-631.]

Please see page 623 for the Editor's Capsule Summary of this article.

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0196-0644/\$-see front matter

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<http://dx.doi.org/10.1016/j.annemergmed.2016.10.008>

INTRODUCTION

Background and Importance

The evaluation of febrile infants in the first 2 months of life is challenging because up to 10% of well-appearing infants can have bacterial infections such as meningitis.¹ Consequently, many febrile infants undergo a lumbar puncture as part of their initial clinical evaluation. Of infants who have a lumbar puncture performed, a substantial proportion (15% to 40%) will have a traumatic lumbar puncture.²⁻⁵ Because introduction of

blood into the cerebrospinal fluid complicates the interpretation of cerebrospinal fluid results, infants with traumatic lumbar punctures have higher rates of hospitalization and are more likely to receive parenteral antibiotics than infants with nontraumatic lumbar punctures.^{2,6,7} Although several methods for correction of the cerebrospinal fluid WBC count for the presence of RBCs have been investigated,⁸⁻¹⁴ these approaches have not been extensively evaluated in the youngest infants and remain controversial.^{4,10}

Editor's Capsule Summary*What is already known on this topic*

Traumatic lumbar punctures are common and difficult to interpret in febrile infants. Correction factors have been advocated but not extensively studied.

What question this study addressed

The authors determined an optimal correction factor from a multicenter retrospective cohort of 20,000 infants younger than 60 days and with lumbar punctures performed. Fourteen percent of lumbar punctures were traumatic; 1.1% had bacterial meningitis.

What this study adds to our knowledge

Using a cerebrospinal fluid RBC:WBC correction ratio of 877:1 would reduce from 78% to 33% the proportion of infants receiving a diagnosis of cerebrospinal fluid pleocytosis while missing 7 cases of bacterial meningitis. The simpler ratio 1,000:1 performed similarly.

How this is relevant to clinical practice

A correction factor of 1,000 RBCs:1 WBC in traumatic lumbar punctures should help identify a low-risk group of infants.

Goals of This Investigation

Our primary objective was to determine the optimal correction factor for cerebrospinal fluid WBC counts in infants aged 60 days or younger and with traumatic lumbar punctures. We derived a traumatic lumbar puncture correction factor to adjust the cerebrospinal fluid WBC count for the presence of cerebrospinal fluid RBCs and compared this derived correction factor, as well as other commonly used correction factors, with the uncorrected cerebrospinal fluid WBC count for the detection of bacterial meningitis.

MATERIALS AND METHODS**Study Design and Setting**

We performed a retrospective cohort study of infants who underwent lumbar punctures at 20 participating centers that contributed cerebrospinal fluid cell counts and culture results for analysis. This study was a planned secondary analysis of the Pediatric Emergency Medicine Clinical Research Committee herpes simplex virus study. The parent herpes simplex virus study sought to determine the incidence of herpes simplex virus disease in infants aged 60 days or younger and undergoing evaluation for

central nervous system infection. Of the 23 pediatric hospitals included in this parent study, we limited our analysis to the 20 sites that contributed cerebrospinal fluid WBC and RBC counts. The study protocol was approved by each of the participating institutional review boards, with permission for data sharing.

Selection of Participants

Infants were eligible for inclusion in our study if they were aged 60 days or younger, presented to a participating emergency department (ED) between January 1, 2005, and December 31, 2013, had a cerebrospinal fluid culture obtained within 24 hours of ED presentation, and had cerebrospinal fluid cell counts available for analysis. We identified eligible infants with electronic search strategies optimized for each study site. Infants with greater than one eligible ED visit in which a lumbar puncture was performed were included for each eligible encounter. Infants without cell counts available were excluded. We defined a traumatic lumbar puncture as a cerebrospinal fluid RBC count greater than or equal to 10,000 cells/mm³ to reflect a level of blood contamination that made the cerebrospinal fluid WBC truly difficult to interpret.^{3,10} We categorized lumbar punctures with a cerebrospinal fluid RBC count less than 10,000 cells/mm³ as nontraumatic.

Methods of Measurement

Data were abstracted electronically at 17 sites and manually at 3. All site abstractors received standardized training through conference calls and biannual in-person meetings.^{15,16} The following data elements were abstracted: demographics (age and sex), visit-level data (study site, date of visit, and disposition), triage temperature, and laboratory data (CBC count with differential, cerebrospinal fluid cell counts, cerebrospinal fluid glucose and protein levels, and cerebrospinal fluid Gram's stain). We manually abstracted the following microbiology results: bacterial cultures (blood, urine, and cerebrospinal fluid), viral cultures (cerebrospinal fluid), and herpes simplex virus polymerase chain reaction cerebrospinal fluid testing.

Outcome Measures

Our primary outcome was a case of bacterial meningitis, defined as an infant with a cerebrospinal fluid culture result (standard or broth) positive for a pathogenic organism.¹⁷ The following bacterial organisms were classified a priori as contaminants: coagulase-negative *Staphylococci*, *Streptococcus viridans*, *Propionibacterium acnes*, and *Corynebacterium* species.

Urinary tract infection was defined as greater than or equal to 50,000 colony forming units/ml or 10,000 to

50,000 colony forming units/ml with an abnormal urinalysis (positive nitrite or pyuria). We defined bacteremia as growth of a bacterial pathogen in blood culture. Herpes simplex virus meningoencephalitis and enteroviral meningitis were defined as a positive cerebrospinal fluid herpes simplex virus and enteroviral polymerase chain reaction test, respectively.

Primary Data Analysis

The unit of analysis was the lumbar puncture. Standard descriptive statistics were used to describe infants with and without a traumatic lumbar puncture. The relationship between cerebrospinal fluid RBCs and cerebrospinal fluid WBCs was explored with linear regression, with cerebrospinal fluid RBC count as the independent variable and cerebrospinal fluid WBC count as the dependent variable, using our a priori understanding of the relation between these cerebrospinal fluid parameters. The slope of the regression line (ie, β -coefficient) was used to derive the cerebrospinal fluid WBC correction factor. We then used a generalized estimating equation to adjust for postnatal age and to cluster by hospital center.

We next evaluated the performance of cerebrospinal fluid pleocytosis for the detection of bacterial meningitis in traumatic lumbar punctures. We defined cerebrospinal fluid pleocytosis as a cerebrospinal fluid WBC count greater than 20 cells/mm³ for infants aged 0 to 28 days and greater than 10 cells/mm³ for infants aged 29 to 60 days.^{6,18} We evaluated the following 5 approaches to interpretation of the cerebrospinal fluid WBC count: (1) uncorrected cerebrospinal fluid WBC count, (2) the regression-based correction factor, (3) a 1,000:1 cerebrospinal fluid RBCs:cerebrospinal fluid WBCs correction factor, (4) a 500:1 cerebrospinal fluid RBCs:cerebrospinal fluid WBCs correction factor, and (5) a correction factor based on using the peripheral RBC to WBC ratio (cerebrospinal fluid WBC corrected=cerebrospinal fluid WBC/[cerebrospinal fluid RBC $\times$ peripheral WBC/peripheral RBC]).⁴ The area under the curve for the receiver operator characteristic curve for cerebrospinal fluid WBC count for the prediction of bacterial meningitis was determined for each of these 5 strategies. For purposes of comparison, we assessed the ability of the cerebrospinal fluid WBC count to detect bacterial meningitis in infants with a nontraumatic lumbar puncture. Given the potential for outpatient management for low-risk febrile infants aged 29 to 60 days, we planned a priori to also report the performance of the 5 approaches to cerebrospinal fluid WBC correction in this subgroup.

Last, we calculated the number needed to treat defined as the number of infants without bacterial meningitis who

would need to be treated for every misclassified infant with bacterial meningitis if we corrected the cerebrospinal fluid WBC count for the presence of cerebrospinal fluid RBCs. This number represents the number of infants without bacterial meningitis who would need to be treated to avoid one misclassified case of bacterial meningitis, using the uncorrected cerebrospinal fluid WBC count instead of the derived correction factor.

Statistical analyses were performed with SPSS (version 23.0; IBM Corporation, Armonk, NY).

RESULTS

Characteristics of Study Subjects

We identified 23,618 lumbar punctures with cerebrospinal fluid culture obtained, of which 20,319 (86%) had cerebrospinal fluid cell counts available. Of these, 14% (2,880/20,319) were traumatic (Figure 1). Among infants with a traumatic lumbar puncture, 1.1% (33/2,880) had bacterial meningitis. We identified the following bacterial meningitis pathogens in infants with traumatic lumbar punctures: Group B *Streptococcus* (8), *Staphylococcus aureus* (8), *Escherichia coli* (7), *Enterococcus* species (3), *Klebsiella* species (2), *Listeria* (1), *Neisseria meningitidis* (1), *Moraxella catarrhalis* (1), *Proteus* species (1), and *Streptococcus pneumoniae* (1). Rates of traumatic lumbar puncture varied significantly by study center (range 6% to 23%). Overall, children with a traumatic lumbar puncture were younger and more likely to be hospitalized (Table 1). Infants aged 29 to 60 days with traumatic lumbar punctures had higher rates of hospitalizations than those with nontraumatic lumbar puncture (87.7%; 1,055/1,203 versus 82.0%; 7,137/8,704; difference 5.7%; 95% confidence interval [CI] 3.5% to 7.7%).

Main Results

Among the 2,880 infants with traumatic lumbar punctures, the median cerebrospinal fluid WBC count was 46 cells/mm³ (interquartile range 17 to 128 cells/mm³) and the cerebrospinal fluid RBC count was 43,050 cells/mm³ (interquartile range 19,050 to 114,397 cells/mm³). The number of cerebrospinal fluid WBCs increased with that of cerebrospinal fluid RBCs in traumatic lumbar punctures (Figure 2). With linear regression, the crude ratio of cerebrospinal fluid RBCs to cerebrospinal fluid WBCs was determined to be 877:1 (95% CI 805 to 961:1), which did not change significantly after adjustment for patient age and hospital center (874:1; 95% CI 803 to 958:1).

When compared with the uncorrected cerebrospinal fluid WBC count, the 877:1 correction factor had lower

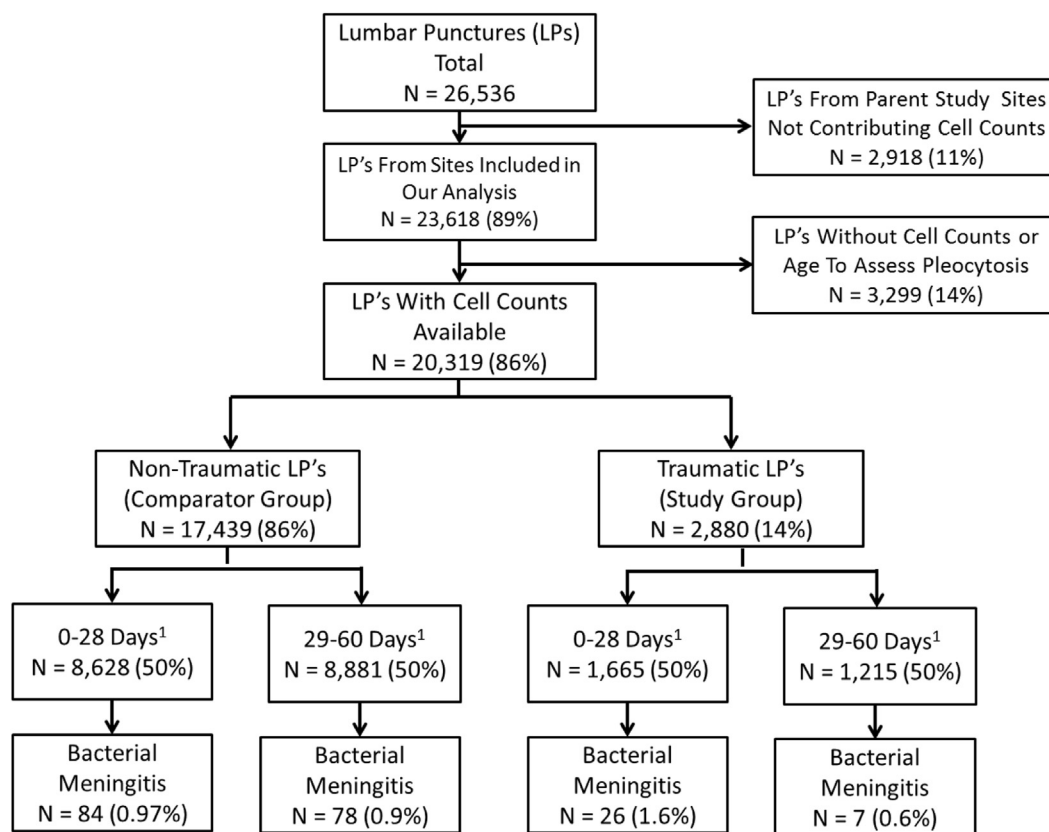


Figure 1. Flow diagram of enrollment.

sensitivity but higher specificity, but the area under the curve for the detection of bacterial meningitis did not differ (Table 2). The derived correction factor classified 45% (1,305/2,880; 95% CI 43% to 47%) more infants with traumatic lumbar punctures as not having cerebrospinal fluid pleocytosis. Among infants aged 29 to 60 days, the derived correction factor had a sensitivity similar to that of the uncorrected cerebrospinal fluid WBC count, but classified 48% (590/1,221; 95% CI 46% to 51%) more infants as not having cerebrospinal fluid pleocytosis (Table 3). Overall and in the subgroup of infants aged 29 to 60 days, the 1,000:1 correction factor performed similarly to the 877:1 correction factor.

Of the 33 infants with a traumatic lumbar puncture and bacterial meningitis, 4 had an uncorrected cerebrospinal fluid WBC count that did not meet the definition of cerebrospinal fluid pleocytosis. All 4 of these infants were aged 0 to 28 days. Application of the derived correction factor misclassified an additional 7 infants with bacterial meningitis (Table 4). Of these, cerebrospinal fluid broth culture from the infant aged 29 to 60 days grew *S aureus*. Because the *S aureus* was considered by the treating clinical team to be a contaminant, the child received only 48 hours of parenteral antibiotics.

For infants aged 28 days or younger, application of the 877:1 correction factor classified 118 of them (95% CI 59 to 253 infants) without bacterial meningitis as not having cerebrospinal fluid pleocytosis for every misclassified infant with bacterial meningitis. For infants aged 29 to 60 days, application of the 877:1 correction factor classified 589 of them (95% CI 106 to 2,439 infants) without bacterial meningitis as not having cerebrospinal fluid pleocytosis for every misclassified infant with bacterial meningitis. The more easily usable 1,000:1 correction factor performed similarly. For infants aged 28 days or younger, application of 1,000:1 correction factor classified 109 of them (95% CI 50 to 233 infants) as not having cerebrospinal fluid pleocytosis for every misclassified case of bacterial meningitis. For infants aged 29 to 60 days, application of a 1,000:1 correction factor classified 532 of them (95% CI 96 to 2,202 infants) as not having cerebrospinal fluid pleocytosis for every misclassified infant with bacterial meningitis. Thus, for every 109 infants younger than 28 days and classified as lacking pleocytosis and for every 532 infants aged 29 to 60 days and classified as lacking pleocytosis, using the 1,000:1 correction, one had bacterial meningitis. In our cohort, for every 223 infants (95% CI 162 to 318 infants) aged 0 to 28 days and 219 infants

Table 1. Comparison between infants with a traumatic and a nontraumatic lumbar puncture.

Characteristic	Nontraumatic LP, N=17,439	Traumatic LP, N=2,880
Demographics		
Age, n/N (%), days*	29.0 (16.0–42.0)	25.0 (13.0–39.0)
0–28	8,628/17,439 (49.5)	1,665/2,880 (57.8)
29–60	8,811/17,439 (50.5)	1,215/2,880 (42.2)
Male infant	9,813/17,439 (56.3)	1,578/2,880 (54.8)
Clinical		
Triage temperature, °C*	99.8 (98.7–100.9)	99.7 (98.7–100.8)
CBC WBC count, cells/mm ³ *	10.4 (7.8–13.5)	10.7 (7.9–13.6)
Admitted, n/N (%)	15,482/17,232 (89.8)	2,682/2,847 (94.2)
Admitted to ICU, n/N (%)	2,529/15,482 (16.3)	394/2,682 (14.7)
Length of stay for admitted, days*	2.2 (1.7–3.0)	2.2 (1.8–3.0)
Microbiology results by age category†		
Urinary tract infection, n/N tested (%), days		
0–29	548/7,911 (6.9)	90/1,519 (5.9)
29–60	702/8,000 (8.8)	84/1,108 (7.6)
Bacteremia, n/N tested (%), days		
0–29	210/8,214 (2.6)	32/1,566 (2.0)
29–60	198/8,318 (2.4)	30/1,144 (2.6)
Bacterial meningitis, n/N (%), days		
0–29	84/8,628 (1.0)	26/1,665 (1.6)
29–60	78/8,811 (0.9)	7/1,215 (0.6)
HSV meningoencephalitis, n/N tested (%), days		
0–29	29/3,532 (0.8)	3/766 (0.4)
29–60	4/1,700 (0.2)	1/306 (0.3)
Enteroviral meningitis, n/N tested (%), days		
0–29	382/1,614 (23.7)	33/293 (11.3)
29–60	440/1,641 (26.8)	35/209 (16.8)

LP, Lumbar puncture; HSV, herpes simplex virus.

*Median (interquartile range).

†Numbers do not sum because microbiologic testing was not available for all patients.

(95% CI 158 to 314 infants) aged 29 to 60 days and with nontraumatic lumbar punctures who were classified as not having cerebrospinal fluid pleocytosis, one had bacterial meningitis.

LIMITATIONS

Our study has limitations. First, many lumbar punctures performed provided adequate sample only for culture and not cell counts and were not included in our analysis. These infants present considerable diagnostic uncertainty for treating clinicians. Second, we selected no upper limit of cerebrospinal fluid RBC counts in our study definition and therefore cannot ensure that the sample contained cerebrospinal fluid and not just peripheral blood. However, the sample was identified as cerebrospinal fluid, representing a clinician's interpretation of the fluid obtained. Third, we used an electronic database and individual medical records were not reviewed and the clinical appearance of study participants could not be assessed. However, ill-appearing infants should be appropriately evaluated and treated for meningitis regardless of their cerebrospinal fluid results.

Fourth, we were unable to determine whether study infants had received antibiotics before diagnostic lumbar puncture, potentially rendering bacterial cultures falsely negative.¹⁹ However, clinicians rarely administer antibiotics to the youngest infants before obtaining a cerebrospinal fluid culture.^{20,21} Additionally, we were unable to determine whether a child with a traumatic lumbar puncture and bacteremia had bacterial meningitis.¹⁷ However, because the rate of bacteremia for study infants was low, significant underreporting of bacterial meningitis is unlikely.²² Fifth, we did not evaluate the performance of cerebrospinal fluid pleocytosis for the detection of viral meningitis because use of viral testing in clinical practice varied substantially. However, viral meningitis is usually a self-limited infection requiring only supportive care. Sixth, we were unable to determine whether all study infants had a fever because only half had a triage temperature greater than 38.0°C (100.4°F). However, infants who were afebrile at triage may have been febrile at home or after initial ED triage. Moreover, the treating clinician deemed it necessary to evaluate all infants for central nervous system infection. Seventh, our study was underpowered to detect small differences in bacterial

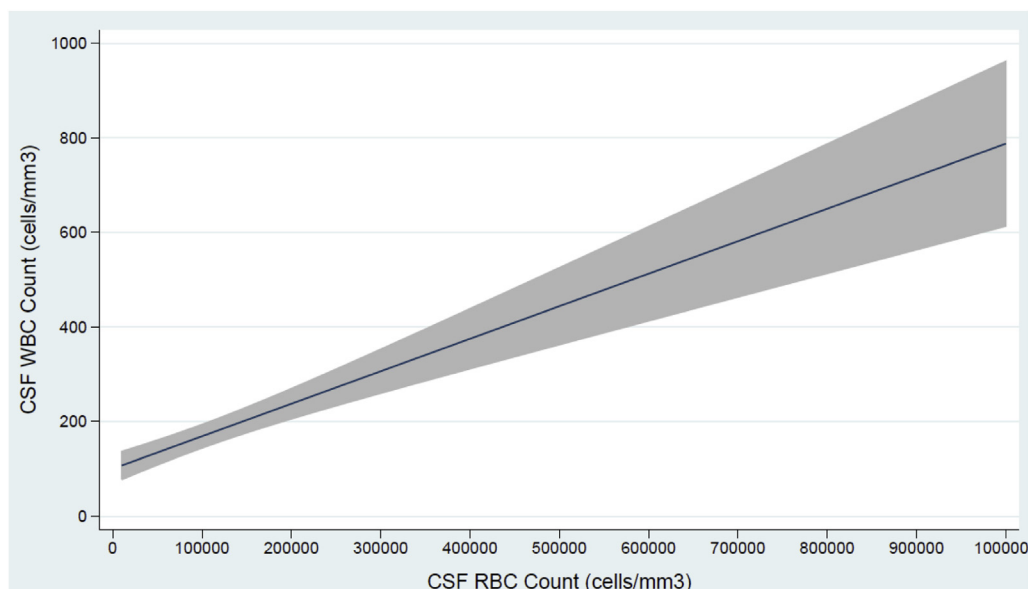


Figure 2. Relationship between cerebrospinal fluid WBC and RBC count in infants with traumatic lumbar punctures. The shaded area represents the 95% CI for estimate of the best fit line. Line derived with linear regression, with CSF WBC count as the dependent variable and CSF RBC count as the independent variable. $\beta=.00114036$, Corresponding to 877 CSF RBCs for every CSF WBC.

meningitis rates because of the small number of cases of bacterial meningitis. However, the rate of bacterial meningitis in our cohort is consistent with what has been reported in current literature and reflects the clinical reality in the era of both widespread conjugate vaccination and maternal antibiotic prophylaxis.²³⁻²⁵ Eighth, we examined only cerebrospinal fluid WBC count as an isolated predictor of bacterial meningitis. Clinicians caring for these infants must consider the cerebrospinal fluid WBC count in the

context of other clinical and laboratory variables when assessing an infant's overall risk of bacterial meningitis.

DISCUSSION

In our large multicenter sample of infants undergoing lumbar puncture as part of their ED evaluation, traumatic lumbar punctures were common. We focused our analysis on infants with bloody spinal fluid (defined as a

Table 2. Comparison of approaches for the detection of bacterial meningitis in all infants aged 60 days or younger days with a traumatic lumbar puncture.

Strategy	Sensitivity, % (95% CI)	Specificity, % (95% CI)	NPV, % (95% CI)	AUC (95% CI)	CSF Pleocytosis, No. (%)	Misclassified Infants With Bacterial Meningitis but No CSF Pleocytosis, N (% of All Infants With Traumatic LP)
Traumatic LPs (N=2,880; 33 bacterial meningitis)						
Uncorrected CSF WBC count	87.9 (71.8–96.6)	21.8 (20.3–23.4)	99.4 (98.4–99.8)	0.71 (0.61–0.81)	2,255 (78)	4 (0.1)
Derived 877:1 correction	66.7 (48.2–82.0)*	67.4 (65.7–69.1)*	99.4 (99.0–99.7)	0.74 (0.63–0.84)	950 (33)*	11 (0.4)
1,000:1 correction	66.7 (48.2–82.0)*	63.4 (61.6–65.2)*	99.4 (98.9–99.7)	0.73 (0.63–0.84)	1,063 (37)*	11 (0.4)
500:1 correction	60.6 (42.1–77.1)*	81.5 (80.0–82.9)*	99.4 (99.1–99.7)	0.75 (0.66–0.85)	547 (19)	13 (0.5)*
Peripheral ratio RBC:WBC†	44.0 (24.4–65.1)*	86.8 (85.4–88.2)*	99.3 (99.0–99.6)	0.70 (0.59–0.81)	318 (13)*	14 (0.6)*
Nontraumatic LPs (N=17,439; 162 bacterial meningitis)						
Uncorrected CSF WBC count	58.6 (50.7–66.3)	85.2 (84.6–85.7)	99.6 (99.4–99.7)	0.77 (0.72–0.81)	2,658 (15)	67 (0.4)

CSF, Cerebrospinal fluid; NPV, negative predictive value; AUC, area under the curve.

*Statistically different from uncorrected CSF WBC at $P<.05$.

†Numbers for peripheral ratio are based on the total number of LPs with peripheral cell counts available, N=2,364.

Table 3. Comparison of approaches for the detection of bacterial meningitis in all infants aged 29 to 60 days and with a traumatic lumbar puncture.

Strategy	Sensitivity, % (95% CI)	Specificity, % (95% CI)	NPV, % (95% CI)	AUC (95% CI)	CSF Pleocytosis, No. (%)	Misclassified Infants With Bacterial Meningitis but No CSF Pleocytosis, No. (% Total)
Traumatic LPs (N=1,221; 7 bacterial meningitis)						
Uncorrected CSF WBC count	100 (59.0–100)	15.2 (13.2–17.3)	100 (98.0–100)	0.79 (0.58–1.00)	1,032 (85)	0
Derived 877:1 correction	85.7 (42.1–99.6)	63.9 (61.3–66.6)*	99.9 (99.3–100)	0.84 (0.65–1.00)*	442 (36)*	1 (0.1)
1,000:1 correction	85.7 (42.1–99.6)	59.2 (56.4–62.0)*	99.9 (99.2–100)	0.87 (0.74–1.00)*	499 (41)*	1 (0.1)
500:1 correction	85.7 (42.1–99.6)	78.7 (76.3–81.0)*	99.9 (99.4–100)*	0.86 (0.70–1.00)*	263 (22)*	1 (0.1)
Peripheral ratio RBC:WBC†	60.0 (14.7–94.7)	85.7 (83.4–87.8)*	99.8 (99.2–100)	0.75 (0.48–1.00)	148 (15)*	2 (0.2)
Nontraumatic LPs (N=8,811; bacterial meningitis 78)						
Uncorrected CSF WBC count	57.7 (46.0–68.8)	82.2 (81.4–83.0)	99.5 (99.4–99.7)	0.74 (0.68–0.81)	1,598 (18)	33 (0.4)

*Statistically different from uncorrected CSF WBC at $P < .05$.

†Numbers for peripheral ratio are based on the total number of LPs with peripheral cell counts available, N=1,023.

cerebrospinal fluid RBC count $\geq 10,000$ cells/mm³) because its interpretation can be the most challenging to clinicians. Application of the derived 877 cerebrospinal fluid RBCs per 1 cerebrospinal fluid WBC correction factor to infants with a traumatic lumbar puncture substantially reduced the number of infants classified as having cerebrospinal fluid pleocytosis while misclassifying only a few cases of bacterial meningitis. **The 1,000:1 to correction factor performed similarly and was considerably easier to apply.** Although cerebrospinal fluid correction factors should be used with caution in the evaluation of the youngest infants, correction of the cerebrospinal fluid WBC count for the presence of peripheral RBCs in older infants has the potential to decrease the number of infants classified as having cerebrospinal fluid pleocytosis without substantial misclassification of infants with bacterial meningitis.

Traumatic lumbar punctures occur commonly in infants,³ yet there is no uniform approach to the

interpretation of cell counts in traumatic cerebrospinal fluid to identify infants at risk of bacterial meningitis.¹⁰ The presence of peripheral blood in the cerebrospinal fluid makes interpretation of cerebrospinal fluid cell counts challenging. Previous studies have investigated methods of correcting cerebrospinal fluid WBC count for the presence of cerebrospinal fluid RBCs in children with a traumatic lumbar puncture.^{4,11,12,14} Heterogeneity of patient populations, various traumatic lumbar puncture definitions, and changes in bacterial meningitis epidemiology because of widespread conjugate vaccination limit the applicability to our study population.^{23,24} The largest study to date included 2,519 infants younger than 30 days with a traumatic lumbar puncture defined as greater than or equal to 500 RBCs/mm³ in the neonatal ICU.⁴ As in our study, correction for a traumatic lumbar puncture using a 1,000:1 correction factor substantially reduced the sensitivity of cerebrospinal fluid WBC count

Table 4. Infants with bacterial meningitis misclassified as not having cerebrospinal fluid pleocytosis by either the 877:1 or the 1,000:1 correction factor.

ID	Age, Days	Sex	Peripheral WBC Count, ×10 ³ /μL	CSF RBC Count/mm ³	Uncorrected CSF WBC Count/mm ³	877:1 Corrected CSF WBC Count/mm ³	1,000:1 Corrected CSF WBC Count/mm ³	CSF Gram's Stain	Bacteremia	CSF Organism
1	6	Female	8.0	38,000	8	0	0	Negative	No	<i>E coli</i>
2	6	Male	18.1	17,360	36	16	19	Negative	No	<i>S aureus</i>
3	15	Female	5.9	110,000	76	0	0	Unknown	No	<i>Enterococcus</i>
4	15	Male	12.0	53,805	53	0	0	Negative	Yes	<i>S aureus</i>
5	16	Male	18.7	36,000	22	0	0	Negative	No	<i>S aureus</i>
6	19	Male	10.2	243,500	50	0	0	Negative	No	<i>E coli</i>
7	23	Male	12.0	77,500	53	0	0	Negative	No	<i>M catarrhalis</i>
8	24	Male	9.1	12,150	17	3	5	Negative	No	<i>E coli</i>
9	24	Male	4.9	16,200	16	0	0	Positive	No	<i>S aureus</i>
10	25	Female	8.4	20,125	14	0	0	Negative	No	<i>Proteus</i>
11	45	Female	11.8	12,900	13	0	0	Negative	No	<i>S aureus</i>

for bacterial meningitis. However, this study did not include infants in the second month of life, in which our correction formula performed with greater accuracy. In a second single-center study of 92 children older than 1 month, cerebrospinal fluid abnormalities associated with bacterial meningitis including cerebrospinal fluid pleocytosis were rarely obscured by the presence of cerebrospinal fluid RBCs.¹³ To our knowledge, our study is the largest to date to evaluate approaches to correction of cerebrospinal fluid WBC count in a large sample of young infants with bloody cerebrospinal fluid.

Consistent with previous work, application of our derived correction factor reduced the number of infants with cerebrospinal fluid pleocytosis at the cost of misclassifying a few additional infants with bacterial meningitis as not having cerebrospinal fluid pleocytosis.⁴ Each of the approaches to correction of the cerebrospinal fluid WBC count for the presence of cerebrospinal fluid RBCs that we evaluated demonstrates the trade-offs between sensitivity and specificity for bacterial meningitis. Because adjustment of cerebrospinal fluid WBC count for the presence of cerebrospinal fluid RBCs reduces the corrected cerebrospinal fluid WBC count, any approach to correction will reduce sensitivity and increase specificity for bacterial meningitis. Compared with the uncorrected cerebrospinal fluid WBC count, our derived 877:1 correction factor missed the fewest cases of bacterial meningitis while appropriately classifying nearly 50% more infants as not having cerebrospinal fluid pleocytosis. However, the 1,000:1 correction factor may be the preferred approach because the correction factor performed similarly and would be considerably easier to apply in clinical practice.

Bacterial meningitis can be challenging to diagnose before the results of bacterial cultures are available, especially in the youngest infants, for whom the clinical examination may not provide adequate discrimination. Even infants with nontraumatic lumbar punctures and normal cerebrospinal fluid WBC counts can have bacterial meningitis.^{4,26,27} The presence of peripheral blood in the cerebrospinal fluid makes clinical decisions about an infant's risk of bacterial meningitis while cerebrospinal fluid cultures are pending even more challenging. Therefore, clinicians must consider cerebrospinal fluid WBC count in the context of the infant's other clinical and laboratory factors. Use of the uncorrected cerebrospinal fluid WBC count will result in increased rates of antibiotic and antiviral exposure, increased hospitalizations, and increased cost for many infants without bacterial meningitis.² Moreover, hospitalization of these infants increases the potential of nosocomial infections, adverse reactions to antibiotics, and

hospital errors.²⁸⁻³⁰ Clinical decisions for an individual patient should be made after the risk of infection is balanced against the risks of hospitalization for the particular infant.

Although most febrile infants younger than 28 days are hospitalized regardless of the results of screening tests, those aged 29 to 60 days may be treated as outpatients if determined to be low risk according to clinical and laboratory criteria.³¹ As in previous work, we observed a significantly higher admission rate for older infants with a traumatic lumbar puncture compared with those without.² Our cerebrospinal fluid WBC count correction factor would allow almost 600 additional infants aged 29 to 60 days and without bacterial meningitis to be appropriately classified as not having cerebrospinal fluid pleocytosis for every misclassified case of bacterial meningitis. Infants aged 29 to 60 days and with traumatic lumbar punctures who lack pleocytosis after correction of their cerebrospinal fluid WBC count may potentially be managed as outpatients. However, given they may still be at risk of other serious bacterial infections, clinicians should strongly consider empiric treatment with a long-acting parenteral antibiotic (eg, ceftriaxone) pending culture results. The one misclassified infant was older than 29 days, grew a likely contaminant from cerebrospinal fluid broth culture only, and was not treated clinically for bacterial meningitis. Therefore, we suggest correction of the cerebrospinal fluid WBC count for the presence of RBCs for otherwise well-appearing infants aged 29 to 60 days and without other findings suggestive of bacterial infection.

In conclusion, in our large multicenter sample of infants aged 60 days or younger and with a traumatic lumbar puncture, application of a correction factor of 1,000 cerebrospinal fluid RBCs per 1 cerebrospinal fluid WBC significantly reduced the number of infants with cerebrospinal fluid pleocytosis while misclassifying 7 additional infants with bacterial meningitis as not having cerebrospinal fluid pleocytosis, of which only 1 was aged between 29 and 60 days. Application of a 1,000:1 cerebrospinal fluid RBC:WBC correction factor in infants aged 29 to 60 days would result in the classification of more than 500 infants without bacterial meningitis as not having cerebrospinal fluid pleocytosis for every infant with bacterial meningitis. Clinicians should consider this trade-off when determining how to interpret cerebrospinal fluid WBC counts in infants aged 29 to 60 days and with traumatic lumbar punctures.

The authors acknowledge the following individuals for their significant contributions as site investigators: Dina Kulik, MD (University of Toronto School of Medicine), Pamela J. Okada,

MD (University of Texas Southwestern Medical Center, Dallas, TX), Alesia H. Fleming, MD, MPH (Emory University School of Medicine, Atlanta, GA), Joseph L. Arms, MD (Children's Hospitals and Clinics of Minnesota, Minneapolis, MN), Aris C. Garro, MD, MPH (Alpert Medical School of Brown University, Providence, RI), Neil G. Uspal, MD (University of Washington School of Medicine, Seattle, WA), Amy D. Thompson, MD (University of Delaware School of Medicine, Wilmington, DE), Paul T. Ishimine, MD (University of California–San Diego School of Medicine, San Diego, CA), Elizabeth R. Alpern, MD (Northwestern University Feinberg School of Medicine, Chicago, IL), Kendra L. Grether-Jones, MD (University of California–Davis School of Medicine, Sacramento, CA), Aaron S. Miller, MD, MSPH (St. Louis University School of Medicine, St. Louis, MO), Jeffrey Louie, MD (University of Minnesota Medical School, Duluth, MN), David Schandower, MD, MPH (Washington University School of Medicine, St. Louis, MO), Sarah J. Curtis, MD, MSc (University of Alberta Faculty of Medicine and Dentistry, Edmonton, AB), Suzanne M. Schmidt, MD (Northwestern University Feinberg School of Medicine, Chicago, IL), and Stuart A. Bradin, DO (University of Michigan Medical School, Ann Arbor, MI).

Supervising editor: Kelly D. Young, MD, MS

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Author contributions: TL and LN conceptualized and designed the study and drafted the article. All authors were responsible for data collection and analysis, approved the final article as submitted, and critically reviewed the article. LN takes responsibility for the paper as a whole.

All authors attest to meeting the four ICMJE.org authorship criteria: (1) Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work; AND (2) Drafting the work or revising it critically for important

intellectual content; AND (3) Final approval of the version to be published; AND (4) Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Funding and support: By Annals policy, all authors are required to disclose any and all commercial, financial, and other relationships in any way related to the subject of this article as per ICMJE conflict of interest guidelines (see www.icmje.org). The authors have stated that no such relationships exist. The network data center managed by the Baylor College of Medicine was supported by the Section of Emergency Medicine, American Academy of Pediatrics and the Baylor College of Medicine. Dr. Lyons was supported by a National Institutes of Health T32 Training Grant (5T32HD040128-12). Dr. Freedman is supported by the Alberta Children's Hospital Foundation's Professorship in Child Health and Wellness. Dr. Balamuth received career development support from NICHD K23-HD082368.

Publication dates: Received for publication May 26, 2016. Revisions received September 13, 2016, and October 3, 2016. Accepted for publication October 5, 2016. Available online December 29, 2016.

Presented at the Pediatric Academic Societies meeting, April 2016, Baltimore, MD.

REFERENCES

1. Baker MD, Avner JR, Bell LM. Failure of infant observation scales in detecting serious illness in febrile, 4- to 8-week-old infants. *Pediatrics*. 1990;85:1040-1043.
2. Pingree EW, Kimia AA, Nigrovic LE. The effect of traumatic lumbar puncture on hospitalization rate for febrile infants 28 to 60 days of age. *Acad Emerg Med*. 2015;22:240-243.
3. Nigrovic LE, Kuppermann N, Neuman MI. Risk factors for traumatic or unsuccessful lumbar punctures in children. *Ann Emerg Med*. 2007;49:762-771.
4. Greenberg RG, Smith PB, Cotten CM, et al. Traumatic lumbar punctures in neonates. *Pediatr Infect Dis J*. 2008;27:1047-1051.
5. Nigrovic LE, Shah SS, Neuman MI. Correction of cerebrospinal fluid protein for the presence of red blood cells in children with a traumatic lumbar puncture. *J Pediatr*. 2011;159:158-159.
6. Baskin MN, O'Rourke EJ, Fleisher GR. Outpatient treatment of febrile infants 28 to 89 days of age with intramuscular administration of ceftriaxone. *J Pediatr*. 1992;120:22-27.
7. Baker MD, Bell LM, Avner JR. Outpatient management without antibiotics of fever in selected infants. *N Engl J Med*. 1993;329:1437-1441.
8. Bonsu BK, Harper MB. Corrections for leukocytes and percent of neutrophils do not match observations in blood-contaminated cerebrospinal fluid and have no value over uncorrected cells for diagnosis. *Pediatr Infect Dis J*. 2006;25:8-11.
9. Osborne JP, Pizer B. Effect on the white cell count of contaminating cerebrospinal fluid with blood. *Arch Dis Child*. 1981;56:400-401.
10. Novak RW. Lack of validity of standard corrections for white blood cell counts of blood-contaminated cerebrospinal fluid in infants. *Am J Clin Pathol*. 1984;82:95-97.
11. Rubenstein JS, Yogev R. What represents pleocytosis in blood-contaminated ("traumatic tap") cerebrospinal fluid in children? *J Pediatr*. 1985;107:249-251.
12. Mayefsky JH, Roghmann KJ. Determination of leukocytosis in traumatic spinal tap specimens. *Am J Med*. 1987;82:1175-1181.

13. Bonadio WA, Smith DS, Goddard S, et al. Distinguishing cerebrospinal fluid abnormalities in children with bacterial meningitis and traumatic lumbar puncture. *J Infect Dis*. 1990;162: 251-254.
14. Mazor SS, McNulty JE, Roosevelt GE. Interpretation of traumatic lumbar punctures: who can go home? *Pediatrics*. 2003;111: 525-528.
15. Gilbert EH, Lowenstein SR, Koziol-McLain J, et al. Chart reviews in emergency medicine research: where are the methods? *Ann Emerg Med*. 1996;27:305-308.
16. Kaji AH, Schrager D, Green S. Looking through the retrospectroscope: reducing bias in emergency medicine chart review studies. *Ann Emerg Med*. 2014;64:292-298.
17. Nigrovic LE, Kuppermann N, Macias CG, et al. Clinical prediction rule for identifying children with cerebrospinal fluid pleocytosis at very low risk of bacterial meningitis. *JAMA*. 2007;297:52-60.
18. Kestenbaum LA, Ebberson J, Zorc JJ, et al. Defining cerebrospinal fluid white blood cell count reference values in neonates and young infants. *Pediatrics*. 2010;125:257-264.
19. Kanegaye JT, Soliemanzadeh P, Bradley JS. Lumbar puncture in pediatric bacterial meningitis: defining the time interval for recovery of cerebrospinal fluid pathogens after parenteral antibiotic pretreatment. *Pediatrics*. 2001;108:1169-1174.
20. Mahajan P, Kuppermann N, Mejias A, et al. Association of RNA biosignatures with bacterial infections in febrile infants aged 60 days or younger. *JAMA*. 2016;316:846-857.
21. Milcent K, Faesch S, Gras-Le Guen C, et al. Use of procalcitonin assays to predict serious bacterial infection in young febrile infants. *JAMA Pediatr*. 2016;170:62-69.
22. Lyons TW, Mcadam AJ, Cohn KA, et al. Impact of in-hospital enteroviral polymerase chain reaction testing on the clinical management of children with meningitis. *J Hosp Med*. 2012;7:517-520.
23. Thigpen MC, Whitney CG, Messonnier NE, et al. Bacterial meningitis in the United States, 1998-2007. *N Engl J Med*. 2011;364: 2016-2025.
24. Poehling KA, Talbot TR, Griffin MR, et al. Invasive pneumococcal disease before and after introduction of pneumococcal conjugate vaccines. *JAMA*. 2006;295:1668-1674.
25. Schrag SJ, Zywicki S, Farley MM, et al. Group B Streptococcal disease in the era of intrapartum antibiotic prophylaxis. *N Engl J Med*. 2000;342:15-20.
26. Smith PB, Cotten CM, Garges HP, et al. A comparison of neonatal Gram-negative rod and Gram-positive cocci meningitis. *J Perinatol*. 2006;26:111-114.
27. Garges HP, Moody MA, Cotten CM, et al. Neonatal meningitis: what is the correlation among cerebrospinal fluid cultures, blood cultures, and cerebrospinal fluid parameters? *Pediatrics*. 2006;117:1094-1100.
28. Walsh KE, Landrigan CP, Adams WG, et al. Effect of computer order entry on prevention of serious medication errors in hospitalized children. *Pediatrics*. 2008;121:e421-e427.
29. DeAngelis C, Joffe A, Wilson M, et al. Iatrogenic risks and financial costs of hospitalizing febrile infants. *Am J Dis Child*. 1983;137:1146-1149.
30. Lieu TA, Baskin MN, Schwartz JS, et al. Clinical and cost-effectiveness of outpatient strategies for management of febrile infants. *Pediatrics*. 1992;89:1135-1144.
31. Aronson PL, Thurm C, Williams DJ, et al. Association of clinical practice guidelines with emergency department management of febrile infants ≤ 56 days of age. *J Hosp Med*. 2015;10:358-365.

IMAGES IN EMERGENCY MEDICINE

(continued from p. 576)

DIAGNOSIS:

Incarcerated paraesophageal hernia. Congenital paraesophageal hernias are rare in children¹⁻⁷; however, if they become incarcerated they can be life threatening. Patients typically present with nonspecific respiratory symptoms and vomiting. This diagnosis is often delayed because of the nonspecific presentation and physicians' inexperience in recognizing this entity. One study found that the average time from onset of symptoms to diagnosis was 5.2 months.² The consequences of delayed diagnosis and treatment can be severe and include incarceration, necrosis, and mortality.

The diagnosis is suggested by the presence of an air- and fluid-filled chest structure, and the criterion standard for diagnosis is an upper gastrointestinal contrast study.

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REFERENCES

1. Baglaj SM, Noblett HR. Paraesophageal hernia in children: familial occurrence and review of the literature. *Pediatr Surg Int*. 1999;15:85-87.
2. Imamoglu M, Cay A, Kosucu P, et al. Congenital paraesophageal hiatal hernia: pitfalls in the diagnosis and treatment. *J Pediatr Surg*. 2005;40:1128-1133.
3. Jawad AJ, Al-Samarrai AI, Al-Mofada S, et al. Congenital para-esophageal hiatal hernia in infancy. *Pediatr Surg Int*. 1998;13:91-94.
4. Jetley NK, Al-Assiri AH, Awadi DA. Congenital para esophageal hernia: a 10 year experience from Saudi Arabia. *Indian J Pediatr*. 2009;76:489-493.
5. Karpelowsky JS, Wieselthaler N, Rode H. Primary paraesophageal hernia in children. *J Pediatr Surg*. 2006;41:1588-1593.
6. Yadav K, Myers NA. Paraesophageal hernia in the neonatal period—another differential diagnosis of esophageal atresia. *Pediatr Surg Int*. 1997;12:420-421.
7. Yazici M, Karaca I, Etensel B, et al. Paraesophageal hiatal hernias in children. *Dis Esophagus*. 2003;16:210-213.